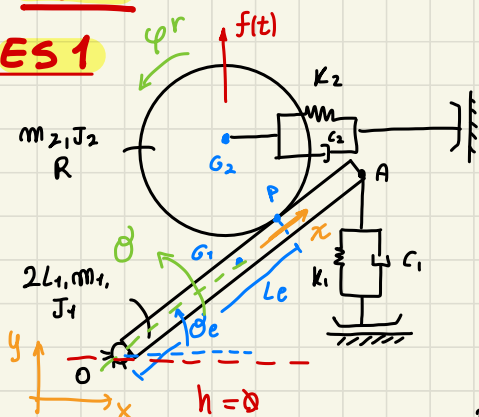




ESE7

ES 1



system

OA rigid bar

and disk rolling without sliding

equilibrium position, bar inclined by θ_e and contact point P @ L_e respect bar

System moves around it!

DOF and set of INDIP COH.

2 bodies: 6 DOF

• HINGED: - 2 D.O.C $\Rightarrow$ 2 DOF

• Rolling wheel - 2 D.O.C

We should find 2 scalar parameters to describe full motion

$$\underline{x} = \begin{Bmatrix} \theta \\ \varphi \end{Bmatrix}$$

GOOD choices

• (motion of bar by θ) & measured from θ_e equilibrium pos

$\theta = 0$ is equilibrium pos so!

and as we can see, if $\theta = 0$ FROZEN and the BAR IS STILL! FROZEN BAR!

rolling of the disk over the bar, φ that if given full still...

we have 3 choices of a good 2 DOF

cohort $\Rightarrow$ we could use

• "x" as displacement of P along the bar (starting @ eq. $x=0$ if $P=L_e$ respect O) (movement of G_2 projection on bar)

• "phi" as disk rotation carefully when using two ROTATIONAL cohort... φ IS ABSOLUTE OR RELATIVE?

φ must be measured respect an abs reference respect ITSELF a ref..

if otherwise φ depends on $\theta + \theta_e$ so it is relative... $\varphi - \theta - \theta_e$ as real angle

HERE we know that DISK ROLLING respect the BAR

" φ^r " (Relative Rotation) and "x" are similar: $x = -R(\varphi^r)$ relationship between the two

while if φ ABSOLUTE respect horizontal axis makes all

trick because $x = -R(\varphi - \theta_e - \theta)$

φ^r RELATIVE ROT.

we prefer to use (φ^r) today!

we prefer solution respect φ^r to link x

to solve the problem we analyze the energetic quantities ...

KINETIC ENERGY

$$T = \frac{1}{2} \dot{\underline{q}}^T [m_{ph}] \dot{\underline{q}}$$

$$\dot{\underline{q}}_m = \begin{bmatrix} v_{G1,x} \\ v_{G1,y} \\ \omega_1 \\ v_{G2,x} \\ v_{G2,y} \\ \omega_2 \end{bmatrix}$$

$$[m_{ph}] = \text{diag} \{ m_1, m_1, J_1, m_2, m_2, J_2 \}$$

(6x6 diag matrix)

relevant
velocity of body 1,2

respect $G_1, G_2 \dots$

sometimes this don't
give contribution on the

don't write it → we can anyway write all components being all

0 contribute in one row of $[\lambda_m]$ all zero!

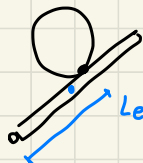
↳ taking that scalar 0 contribute HERE both x,y of the bodies and $\omega_{1,2}$ are important!

$$[\lambda_m] = \begin{bmatrix} -L_1 \sin(\theta_e) & 0 & 0 & 0 & 0 & 0 \\ L_1 \cos(\theta_e) & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ - (L_1 \sin(\theta_e) + R \cos(\theta_e)) & -R \cos(\theta_e) & 0 & 0 & 0 & 0 \\ L_1 \cos(\theta_e) - R \sin(\theta_e) & -R \sin(\theta_e) & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_{G1,x} \\ v_{G1,y} \\ \omega_1 \\ v_{G2,x} \\ v_{G2,y} \\ \omega_2 \end{bmatrix}$$

• IF $\dot{\varphi}^r = 0, \ddot{\varphi} \neq 0$

(FIRST column)

Rotating around
O by θ_1 (NOT depending by
our choice of θ , measure)
We write terms
around equilibrium,
NOT in a general
motion
 $\omega_1 = \dot{\theta}$



Carefully that φ^r
"RELATIVE" is relevant!

it affects also when
 $\varphi^r = 0$! freezing $\dot{\varphi}^r = 0$
(Relative rotation)

IF $\varphi^{\text{ABSOLUTE}} = 0$ frozen

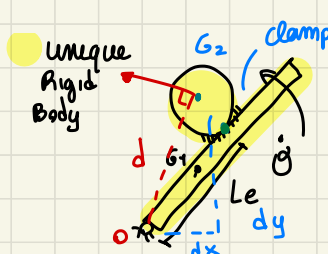
all bodies → Translatory motion so all

moving as P on BAR IF here is best to choose a

Relative rotation (φ^{rel}) → HARD PART...

Remember! When freezing Relative Rotation, DISK will not perform TRANSLATORY
MOTION → it has same rotation as body 1 and because on P

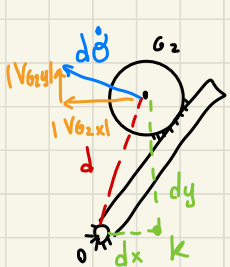
DISK and BAR has same velocity (NO SLIDING) → they have one common
point, they are clamped forming one rigid body



clamp the disk with BAR

to find G_2 velocity → RIVALL'S THEOREM holds, so
 v_{G2} perpendicular to OG_2 with magnitude

$d \cdot \dot{\theta} = |\vec{v}_{G2}|$ BUT we should find
d as respect dx and dy ⇒ TRICK!



IF we represent $d \cdot \dot{\theta}$ horiz/vertical

$|V_{G2x}|, |V_{G2y}|$ JUST amplitude

you can see clearly how

OK $G_2 \triangleq$ to velocity vector $|V_{G2x}|, |V_{G2y}|, d \dot{\theta}$
similar
 $\downarrow$

$$\begin{cases} |V_{G2x}| = d \dot{\theta} \cdot \frac{dy}{d} = \dot{\theta} dy & \text{horiz comp of } G_2 \text{ velocity has} \\ & \text{magnitude respect } \dot{\theta} \cdot \text{vertical dist} \\ |V_{G2y}| = d \dot{\theta} \cdot \frac{dx}{d} = \dot{\theta} dx & \text{vertical vel of } G_2 \text{ has mag} \\ & \text{respect } \dot{\theta} \cdot \text{horiz dist} \end{cases}$$

so on the table -- [1m] @ equilibrium

$$|OG_2|_x = L \cos(\theta_e) - R \sin(\theta_e) \rightarrow |V_{G2y}| = \dot{\theta} |OG_2|_x \quad \dot{\theta} > 0 \rightarrow V_{G2y} > 0 \quad (+)$$

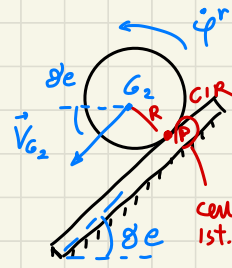
$$|OG_2|_y = (L \sin(\theta_e) + R \cos(\theta_e)) \rightarrow \dot{\theta} > 0 \Rightarrow V_{G2x} < 0 \quad \text{with } (-)$$

even if $\dot{\varphi}^r = 0 \rightarrow$ disk welded to bar

Rotating with $\dot{\theta}$ angular speed, $\omega_2 = 1 \cdot \dot{\theta}$ because φ^r

$\dot{\theta} = 0, \dot{\varphi}^r \neq 0$

easy to see, bar still while respect disk, $\dot{\varphi}^r$, disk ROLL DOWN the bar



$V_{G2} \parallel$ BAR inclination θ_e

With magnitude

$$|V_{G2}| = R \dot{\varphi}^r$$

and project it on x,y respect θ_e

center of 1st. rotation

and on this case $\omega_2 = \dot{\varphi}^r \cdot 1$
 absolute angular speed

POTENTIAL ENERGY { elastic
gravitational

$$V_{el} = \frac{1}{2} \underline{y}_k^T [k_{ph}] \underline{y}_k \quad \underline{y}_k = \begin{Bmatrix} \Delta e_1 \\ \Delta e_2 \end{Bmatrix} \quad [k_{ph}] = \text{diag}\{k_1, k_2\} = \begin{bmatrix} k_1 & 0 \\ 0 & k_2 \end{bmatrix}$$

Jacobian to Link:

$$[\lambda_k] = \begin{bmatrix} 2L_1 \cos(\theta_c) & 0 \\ L_1 \sin(\theta_c) + R \cos(\theta_c) & R \cos(\theta_c) \end{bmatrix} \begin{Bmatrix} \Delta e_1 \\ \Delta e_2 \end{Bmatrix}$$

@ equilibrium position

here interested on I ORD terms
(linearized relationship!)

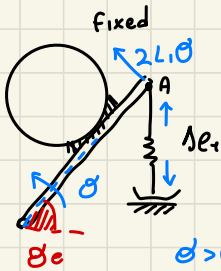
$\Delta e_1, \Delta e_2$ respect infinitesimal
displacement to II ORD is
negligible!

↳ consider
later (NOT
MATRIX approach)

CONVENTION
← mm → (+)

• $\theta \neq 0, \varphi = 0$

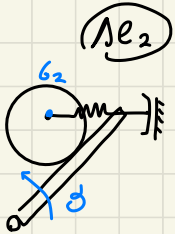
single rigid body weld



(Δe_1) related to y_A ... vertical displacement of point A.
we can use RIVALS linking Rot/disp. in some way
of linking v/ω ...

$$\frac{dx}{dt} / \frac{d\theta}{dt} \rightarrow \text{some relation.}$$

$\theta > 0$ increase
 Δe_1



horiz. component of G_2 with (-) sign

$G_{2x} > 0$ means compression

($\Delta e_2 = -G_{2x}$) $\Rightarrow$ small infinitesimal displacement,
same relationship respect Rotation speed

some relation

$\dot{\theta} / V_{G_{2x}}$ but negative (-)

• $\varphi \neq 0, \theta = 0$ disk rolling

obw/m from BAR FIXED $\rightarrow \Delta e_1 = R$

while (+) elongation of Δe_2
by: $R \cos(\theta_c)$

as $\dot{\varphi} R / V_{G_{2x}}$
but with (-) sign

II ORD STIFFNESS TERMS: 2nd ORD elastic stiffness matrix

we have to write NONLINEAR EXPRESSION linking elongation of $\Delta l_1, \Delta l_2$ respect θ, φ

$$\Delta l_1 = \underbrace{2L_1 \sin(\theta + \theta_e)}_{\substack{\text{vertical position} \\ \text{of point A II CHOICES.}}} - \underbrace{2L_1 \sin(\theta_e)}_{\substack{\text{equilibrium} \\ \text{CONSTANT TERMS}}} + \Delta l_{10} =$$

= I) for this you can use POWER SERIES EXPANSION

$\sin(\theta_e + \theta) \big|_{\theta=0}$ and expand $2L_1 \sin(\theta + \theta_e) + \frac{\partial^2}{\partial \theta^2} \big|_{\theta=0} + \frac{(\partial^2)}{(\partial \theta^2)} \big|_{\theta=0}$ 2nd ORD $\rightarrow$ easy if JUST depends on θ , if depends only of φ we need more derivatives

II) because spring

elongation depends on θ only, we have a simple way (instead of $\Delta l_1(\theta, \varphi)$ it is complex) we develop $\sin(\theta_e + \theta)$ using usual formulae of expansion $\sin(\theta) \sim \theta$
 $\cos(\theta) \sim 1 - \theta^2/2$ } II ORD expansion near $\theta = 0$

$$= 2L_1 \cos(\theta_e) \underbrace{\sin(\theta)}_{\approx \theta} + 2L_1 \sin(\theta_e) \underbrace{\cos(\theta)}_{\approx (1 - \theta^2/2)} =$$

$$= \underbrace{2L_1 \cos(\theta_e) \theta}_{\substack{\text{FIRST ORDER} \\ \text{PART of Relation,}}} - \underbrace{2L_1 \sin(\theta_e) \frac{\theta^2}{2}}_{\substack{\text{2nd ORD} \\ \text{TERMS}}} \quad \text{CONSTANT TERM, negligible}$$

CONSISTENT with what we found previously $\rightarrow$ you can also neglect the $[1_k]$ creation and check for Jacobian terms directly here!

$$\frac{\partial^2 \Delta l_1}{\partial \theta^2} \bigg|_{\theta=0} \quad \frac{\partial^2 \Delta l_1}{\partial \varphi^2} \bigg|_{\theta=0} = 0$$

$$[K_{el, II, 1}] = K_1 \Delta l_{01} \begin{bmatrix} -2L_1 \sin(\theta_e) \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} \frac{\partial \Delta l_1}{\partial \theta \partial \varphi} \\ 0 \\ 0 \end{bmatrix}$$

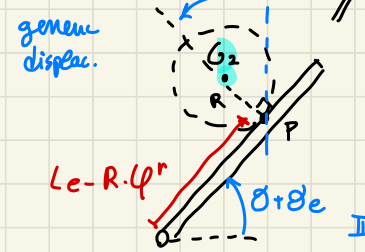
STATIC FORCE @ equilibrium

multiply $\varphi \cdot \theta$

multiply $\varphi^2/2$

coeff multiplying $\theta^2/2$

developing total $\Delta \ell_2 = -X_{G_2} = -[(L_e - R\varphi^r) \cos(\vartheta_e + \vartheta) - R \sin(\vartheta_e + \vartheta)] + \text{const} =$
 (+ constant terms: relative to $\Delta \ell_2$ an equilibrium position)



disk rolling over bar of $\varphi^r \dots$
 so P moved along bar by $R\varphi^r$

$$= -(L_e - R\varphi^r) \left(\cos(\vartheta_e) \left(1 - \frac{\vartheta^2}{2}\right) - \sin(\vartheta_e) \cdot \vartheta \right) + R \cos(\vartheta_e) \vartheta + R \sin(\vartheta_e) \left(1 - \frac{\vartheta^2}{2}\right) =$$

$$= -\cancel{L_e \cos(\vartheta_e)} + R \varphi^r \cos(\vartheta_e) + \cancel{L_e \cos(\vartheta_e)} \frac{\vartheta^2}{2} + \dots \text{higher order terms } (\varphi \frac{\vartheta^2}{2})$$

const I ORD, match $0m[\lambda_k]$ 2nd ORD TERMS $\vartheta^2/2$

$$+ L_e \sin(\vartheta_e) \cdot \vartheta - R \varphi^r \sin(\vartheta_e) \vartheta + R \cos(\vartheta_e) \vartheta + R \sin(\vartheta_e) \vartheta - \cancel{R \sin(\vartheta_e)} \frac{\vartheta^2}{2}$$

terms multiply $\varphi \cdot \vartheta \rightarrow -R \sin(\vartheta_e)$ NO COEFF respect $\varphi^2/2$

$[L_e \sin(\vartheta_e) + R \cos(\vartheta_e)] 0m[\lambda_k]$

from the expression!

$$\Rightarrow [K_{\ell II, 2}] = k_2 \Delta \ell_{02} \begin{bmatrix} L_e \cos(\vartheta_e) - R \sin \vartheta_e & -R \sin(\vartheta_e) \\ -R \sin(\vartheta_e) & 0 \end{bmatrix}$$

GRAVITATIONAL, we should do the same thing for h_{G1}, h_{G2} taking $[K_g]$

$$V_g = m_1 g h_1 + m_2 g h_2 \quad \begin{matrix} h_1 = h_{G1} \\ h_2 = h_{G2} \end{matrix} \quad \begin{matrix} \text{height over fixed reference} \\ \text{as the HINGE O} \end{matrix}$$

$$\bullet h_1 = L_1 \sin(\vartheta_e + \vartheta) =$$

$$\simeq \cancel{L_1 \cos(\vartheta_e)} \cdot \vartheta + L_1 \sin(\vartheta_e) \left(-L_1 \sin(\vartheta_e) \right) \frac{\vartheta^2}{2}$$

I ORD, going to cancel out with elastic pot en. I ORD! II ORD terms respect $\frac{\partial^2 h_1}{\partial \vartheta^2} \Big|_{\vartheta=0}$

$$[K_{g1}] = m_1 g \begin{bmatrix} -L_1 \sin(\vartheta_e) & 0 \\ 0 & 0 \end{bmatrix}$$

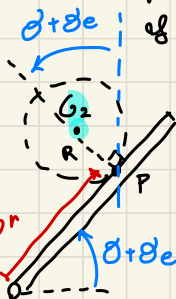
• $h_2 = (L_e - R\varphi) \sin(\vartheta_e + \vartheta) + R \cos(\vartheta_e + \vartheta) = (L_e - R\varphi) (\cos(\vartheta_e)\vartheta + \sin(\vartheta_e)(1 - \frac{\vartheta^2}{2}) +$

more complex situation

$h_2 = y_{G_2} :$

projection along y

$+ R (\cos(\vartheta_e)(1 - \frac{\vartheta^2}{2}) - \sin(\vartheta_e)\vartheta) =$



$= L_e \cos(\vartheta_e)\vartheta + L_e \sin(\vartheta_e) \left(-L_e \sin(\vartheta_e) \frac{\vartheta^2}{2} \right)$

$- R \cos(\vartheta_e) \varphi \cdot \vartheta - R \sin(\vartheta_e) \cdot \varphi + (\varphi \cdot \vartheta^2) \text{ 3rd ORD... + higher...}$

$+ R \cos(\vartheta_e) \left(-R \cos(\vartheta_e) \frac{\vartheta^2}{2} \right) - R \sin(\vartheta_e) \vartheta$

In the end we obtain Relevant terms leading to:

$[K_{g_2}] = m_{g_2} \begin{bmatrix} -(L_e \sin(\vartheta_e) + R \cos(\vartheta_e)) & -R \cos(\vartheta_e) \\ -R \cos(\vartheta_e) & 0 \end{bmatrix}$

VIRTUAL

• WORK respect the $f(t)$ acting vertically on G_2

$\delta W = \delta y_F^T \cdot \underline{f}_{ph}$

↳ here $\underline{f}_{ph} = \{ f(t) \}$

So consistently

$\begin{pmatrix} \delta \vartheta \\ \delta \varphi^r \end{pmatrix}$

$\delta y_F = \{ \delta y_{G_2} \} = [\lambda_F] \cdot \begin{pmatrix} \delta \vartheta \\ \delta \varphi^r \end{pmatrix}$

$[\lambda_F] = [L \cos(\vartheta_e) - R \sin(\vartheta_e) \quad -R \sin(\vartheta_e)] \delta y_{G_2}$

↑
JUST ONE
TERM, VIRTUAL
displacement of G_2

I ORD relation between displacement /

variation $\rightarrow$ same as horizontal velocity V_{G_2y} respect $\dot{\vartheta}$, $\dot{\varphi}^r$

$V_{G_2y} [L \cos(\vartheta_e) - R \sin(\vartheta_e) \quad -R \sin(\vartheta_e)]$

STOP $\rightarrow$ from here we can WORK on MATLAB to solve the problem!